Developer Pack

Ledgerline

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Author	Ledgerline platform team

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01

API Reference

API Reference

Overview

Base URL	`http://localhost:8000` (`Dockerfile` , `docker-compose.yml`)
Format	JSON
Source	Route detection in `app/main.py` and `app/routers/orders.py` (medium confidence), confirmed by reading the code
Version	0.9.2 (`FastAPI(version=...)` in `app/main.py`)

FastAPI also serves the generated OpenAPI schema at `/openapi.json` and interactive docs at `/docs` untested.

Authentication

Every `/orders` endpoint requires an API key in the `x-api-key` header (`dependencies=[Depends(require_api_key)]` in `app/routers/orders.py`). The key is compared with the `API_KEY` environment variable (`app/auth.py`).

```
x-api-key: <your API key>
```

Situation	Response
Header missing	`422` validation error (header is required)
Wrong key	`401` `{"detail": "invalid API key"}`
`API_KEY` not set on the server	`500` : `os.environ["API_KEY"]` raises `KeyError`

Endpoints

Method	Path	Summary	Auth
`GET`	`/health`	Liveness check	no
`GET`	`/orders`	List orders	yes
`POST`	`/orders`	Create an order	yes
`GET`	`/orders/{order_id}`	Get one order	yes
`DELETE`	`/orders/{order_id}`	Cancel an order	yes

`GET /health`

Returns service status (`app/main.py`).

```
curl http://localhost:8000/health
```

Response `200`

```
{"status": "ok"}
```

`GET /orders`

Lists orders (`list_orders` , `app/routers/orders.py`).

Name	In	Type	Required	Description
`limit`	query	int	no	Maximum results, default `50`

```
curl "http://localhost:8000/orders?limit=10" -H "x-api-key: $API_KEY"
```

Response `200` : an array of `OrderOut` . The current implementation always returns `[]` .

`POST /orders`

Creates an order from an `OrderIn` body (`app/schemas.py`).

```
curl -X POST http://localhost:8000/orders \
  -H "x-api-key: $API_KEY" -H "Content-Type: application/json" \
  -d '{"customer_id": 42, "sku": "MUG-001", "quantity": 12}'
```

Warning

Not implemented yet: the handler raises `501 {"detail": "not implemented"}` . The declared success status is `201` .

`GET /orders/{order_id}`

Name	In	Type	Required	Description
`order_id`	path	int	yes	Order ID

The current handler always returns `404 {"detail": "order not found"}` .

`DELETE /orders/{order_id}`

Cancels an order. Returns `204 No Content` .

```
curl -X DELETE http://localhost:8000/orders/42 -H "x-api-key: $API_KEY"
```

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Schemas

OrderIn

Field	Type	Required
`customer_id`	int	yes
`sku`	string	yes
`quantity`	int	yes

OrderOut: every `OrderIn` field, plus:

Field	Type	Notes
`id`	int	Order ID
`status`	string	`pending` by default in the database (`app/models.py`)

Error codes

Status	Detail	When
401	`invalid API key`	`x-api-key` doesn't match `API_KEY`
404	`order not found`	`GET /orders/{order_id}`
422	validation error	missing header, or a body or path value of the wrong type
501	`not implemented`	`POST /orders`

02

Data Model

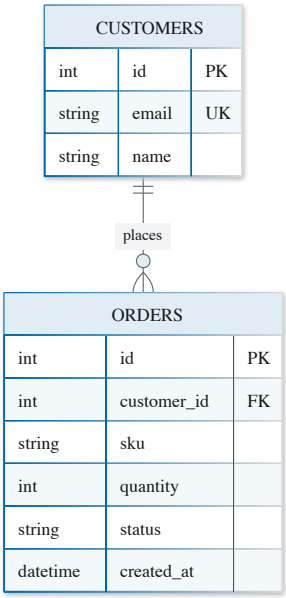
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Data Model

Overview

Database	PostgreSQL 16 (`postgres:16` in `docker-compose.yml`)
Access	SQLAlchemy 2.0 ORM, typed `Mapped[...]` models
Source	`app/models.py` (models), `alembic/versions/` (migrations)
Entities	2: `Customer` , `Order`

Entity relationship diagram



Entities

Customer

A wholesale buyer (`app/models.py` , table `customers`).

Field	Type	Null	Key	Notes
`id`	int	no	PK	
`email`	varchar(255)	no	unique	
`name`	text	no		

Order

An order line for one SKU (`app/models.py` , table `orders`).

02

Data Model

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Field	Type	Null	Key	Notes
`id`	int	no	PK	
`customer_id`	int	no	FK → `customers.id`	indexed
`sku`	varchar(64)	no		
`quantity`	int	no		
`status`	text	no		default `pending`
`created_at`	timestamp	no		default `datetime.utcnow`

Relationships

From	To	Cardinality	On delete	Evidence
Customer	Order	1:N	database default assumption	`relationship(back_populates=...)` in `app/models.py`

Indexes and constraints

Table	Index / constraint	Columns	Type
`customers`	primary key	`id`	btree
`customers`	unique	`email`	btree
`orders`	primary key	`id`	btree
`orders`	index	`customer_id`	btree
`orders`	foreign key	`customer_id` → `customers.id`	constraint

Migrations

- Tool: Alembic · Location: `alembic/versions/` · Count: 2
- `0001_init.py` : create customers and orders
- `0002_order_status.py` : add order status

```
make migrate      # alembic upgrade head
```

Note

`created_at` uses `datetime.utcnow`, which returns naive timestamps. Store and compare times as UTC throughout.

03

Assumptions and open questions

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Assumptions and open questions

Not included: Developer Guide, Architecture, Contributing Guide, Testing Strategy, CI/CD and Deployment, Environment and Configuration Reference, Security Threat Model, Dependency Inventory, Tech Debt Register, Integration Guide (not selected for this sample).

#	Chapter	Item	Type	Why unverified	Who can confirm
A1	API Reference	`/docs` and `/openapi.json` are reachable	assumption	FastAPI default, not run	backend lead
Q1	API Reference	When will `POST /orders` and `GET /orders/{order_id}` be implemented?	open question	handlers are stubs	backend lead
Q2	API Reference	Should a missing `API_KEY` fail at startup instead of per request?	open question	design choice	backend lead
A2	Data Model	On-delete behaviour for orders	assumption	no `ondelete=` set in the model	backend lead
Q3	Data Model	Should `status` be an enum?	open question	free text today	backend lead